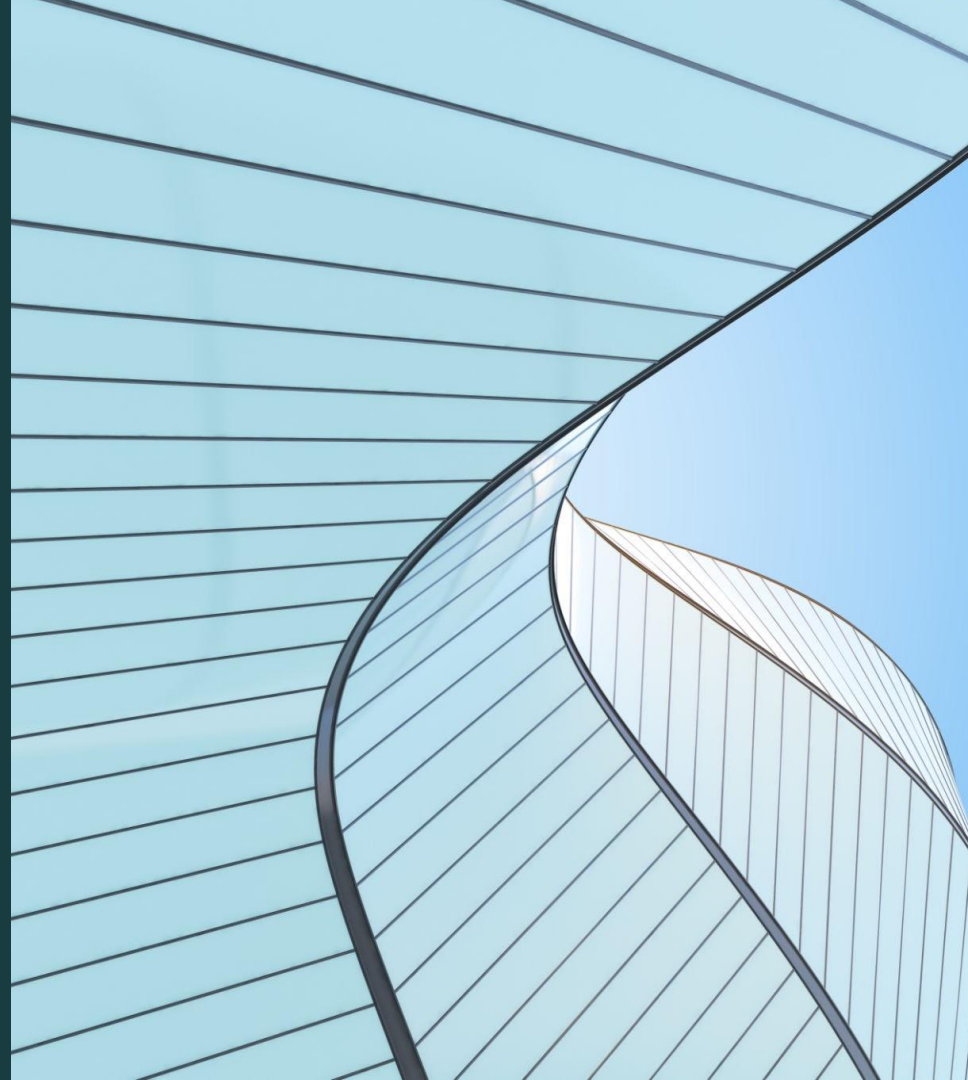


QuantaFlex ETF

MSDS 451
Final Term Project



Project Overview

Project idea: A rules-based, actively managed multi-asset ETF

Goal: Combine diversification, transparency, and systematic decision rules

Main asset classes:

- U.S. growth equity
- investment-grade bonds
- global equities
- commodities
- cash



Investment Philosophy and Assets

Philosophy:

- diversification
- rules-based allocation
- momentum and trend awareness
- volatility-sensitive risk control

Assets:

- NVDA
- AGG
- ACWI
- GLD
- DBC
- BIL
- SPY benchmark



Data and Methods

Data source: Yahoo Finance via yfinance

Period: 2008–2024

Frequency: daily

Return measure: daily log returns

Baseline model: equal-weight long-only buy-and-hold

Testing:

- benchmark comparison
- Monte Carlo simulation
- fee consideration



Performance Results

Metric	QuantaFlex	SPY
Annualized Return (%)	8.03	10.85
Annualized Volatility (%)	12.92	19.95
Sharpe Ratio	0.62	0.54
Max Drawdown (%)	-39.54	-51.48
Alpha (%)	2.64	—
Beta	0.50	—



What the Results Mean

QuantaFlex had:

- lower return than SPY
- lower volatility
- lower drawdown
- slightly better Sharpe ratio

Interpretation:

- stronger as a defensive diversified portfolio
- not yet strong enough as a pure active return product



Monte Carlo and Fees

Monte Carlo results:

- mean 5-year ending value ≈ 1.55
- median 5-year ending value ≈ 1.50

Fee takeaway:

- modest fees may be acceptable
- stronger active rules are needed to justify a full fund launch

Management Recommendation

Investor view:

- promising for lower-risk diversified exposure

Business view:

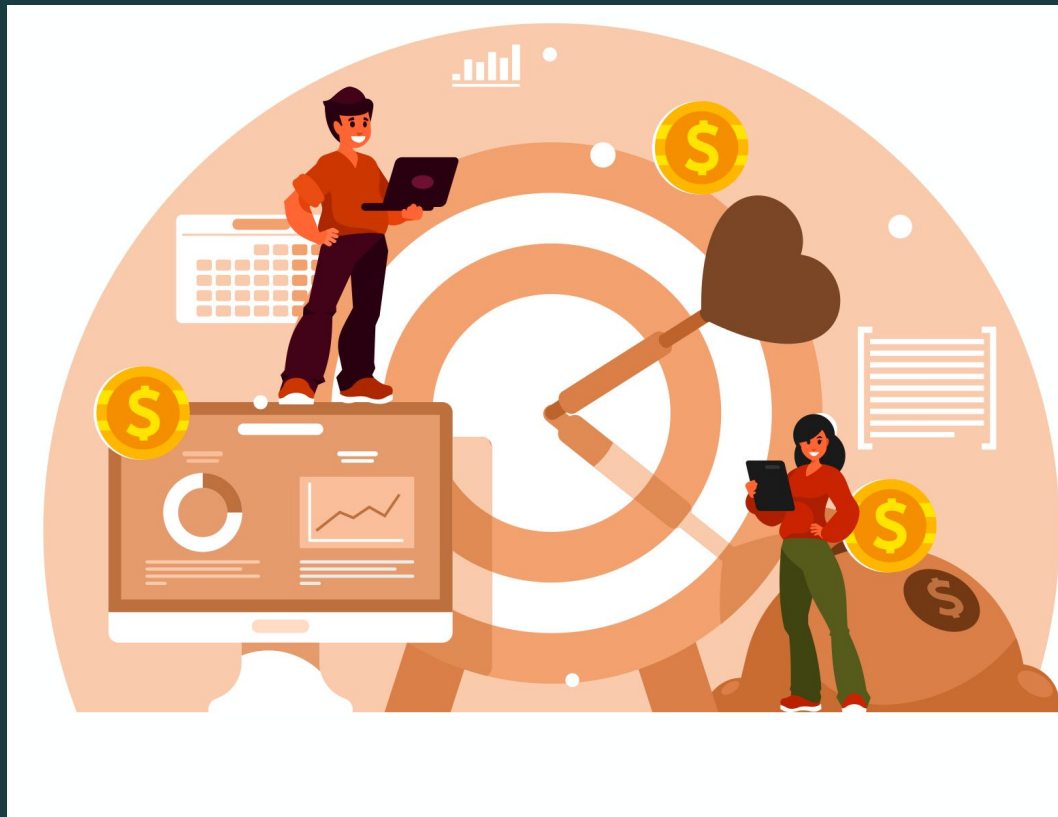
- not ready to launch yet

My role:

- researcher / quantitative strategist

Next step:

- test active rules and post-fee performance



Conclusion

QuantaFlex is:

- feasible
- transparent
- diversified
- supported by data and simulation

Final conclusion:

- good ETF concept
- needs stronger active-rule validation before commercialization

